
AERIAL VIEW OBJECT CLASSIFICATION ON THE BIRDSAI DATASET

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ABSTRACT

With increased poacher activity targeting rhinos and elephants, conservation organizations are left responsible for monitoring large swaths of land for unauthorized human presence. The current system for surveillance of this land is manual inspection of long-wave infrared (LWIR) video feed captured by UAVs. This solution, while functional, is tedious and is subject to human error. We propose a real-time computer vision monitoring system based on a pre-trained YOLOv5 model to distinguish between humans and animals in LWIR imagery. By combining pseudo-labeling of synthetic data with weighted cross-entropy loss, our approach was able to achieve 76% recall on the human class, a steep improvement over the 39% achieved from training on just real images.

1 Background/Theory

A similar project was conducted by Bondi et al. [1] in 2018. Their project utilized an RCNN model, opting for the improved accuracy of this model over other potential architectures. In field testing, their model was able to achieve approximately 37% recall in the human class, which was a significant improvement over the current human monitoring system. We hope to build off of this project using a YOLO architecture for increased inference speed.

The YOLO architecture we use for our model was first described by Redmon et al.[2] in 2015. These models frame object detection problems as a regression problem, allowing for inference in a single-pass of the image. This speeds up the model, which is critical for our model to perform real-time analysis of LWIR video feed.

The YOLOv5 model specifically is constructed with a CSPDarkNet-53 backbone, which is a 53-layer convolutional neural network. CSPNets were first introduced by Wang et al[3]. CSPNet splits the input feature map x_0 into two parts:

$$x_0 = [x'_0, x''_0]$$

- Path 1 (x''_0) passes through the network
- Path 2 (x'_0) bypasses the network and is fused afterwards

This increases gradient diversity, as well as reduces the computational load of training the model. YOLOv5 models also make use of anchor boxes described by Zhong et al[4], which are a set of hypothetical bounding boxes produced during training that represent the different sizes of objects present in the training set. This allows the model to perform a refinement problem at inference time rather than forming new bounding boxes for each image, increasing the accuracy of the model.

2 Data

The BIRDSAI [5] dataset is composed of 27000 LWIR images of the African savannah, of which only 447 are real and labeled. The remainder are synthetically generated and unlabeled.



Figure 1: An example of the real, labeled data provided

In order to generate more training data, we applied an initial YOLOv5 detector that was trained exclusively on the 447 real images to a randomly selected subset of 553 synthetic images. After manual verification of these pseudo-labels, we assembled a 1000 image labeled set. Our dataset has two issues that we encountered through our training process. For one, there is a severe class imbalance. In our newly generated 1000 image dataset, there are only 280 instances of the human class, as compared to 4503 instances of the animal class. This issue was somewhat addressed in the generation of our pseudo-labels, as the original dataset only had 140 human instances, but this imbalance still provides an issue in training our model. Secondly, the images in our dataset are taken from a variety of vantage points.



Figure 2: An example of an image in our dataset taken from a high vantage point

As can be clearly seen in comparing the two provided images, despite both images containing members of the animal class, the appearance, most importantly the size, of the instances look quite different depending on the vantage point. This difference requires our model to generalize well to the appearance of each class from different perspectives.

3 Methodology

We adopted a pretrained YOLOv5 with a CSPDarknet-53 backbone. This choice was made for its favorable inference speed while still maintaining a good level of accuracy. We trained our model for 10 epochs.

3.1 Loss Function

Given the class imbalance, we applied a weighted cross-entropy loss defined as

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [w_{human} y_i \log p_i + w_{animal} (1 - y_i) \log (1 - p_i)] \quad (1)$$

where w_{human} and w_{animal} are the weights assigned to each class. We set these weights to $w_{human} = 16$ and $w_{animal} = 1$, as the ratio between animal and human instances in the training set is 16:1

3.2 Optimizer

We trained the model using the Adam optimizer with momentum = 0.937 and weight decay = 5×10^{-5}

4 Results/Analysis

4.1 Initial Model

Our initial model was trained only on the 447 real images provided in the BIRDSAI dataset. This model was then applied to an unseen test set, which was made up of 200 hand-labeled images that were randomly selected from the synthetic data provided. This resulted in the confusion matrix below

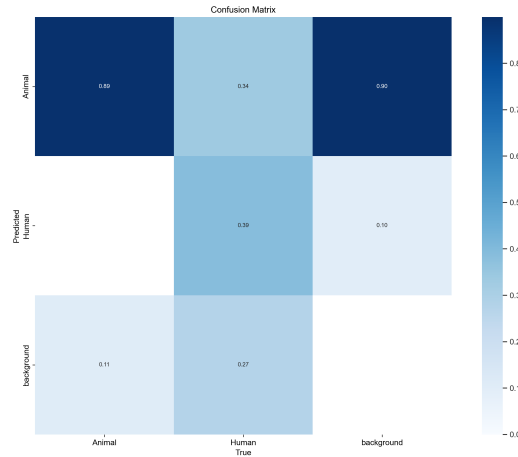


Figure 3: Confusion Matrix of initial YOLOv5 model

The metric that we are most concerned with is recall for the human class, as in the real-life application of this model, catching all instances of humans in the video footage is more important than preventing an occasional misclassification. As can be seen in the confusion matrix given, the initial model that we trained had relatively poor recall for the human class at 39%. However, this model was useful in generating pseudo-labels for the 553 synthetic images that we introduced into our training set

4.2 Model Trained with Pseudo-labeled Data

The YOLOv5 model was then re-trained with the same hyperparameters on this 1000 image dataset, which resulted in the confusion matrix below.

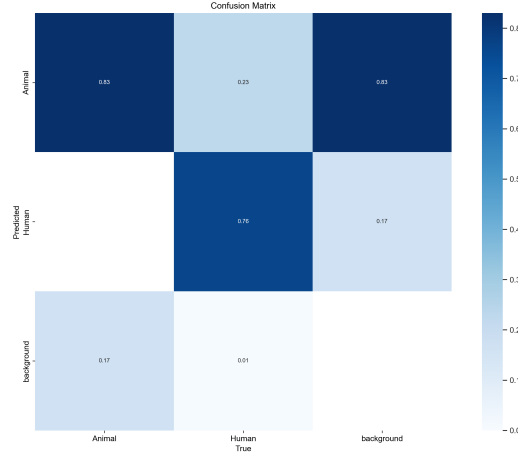
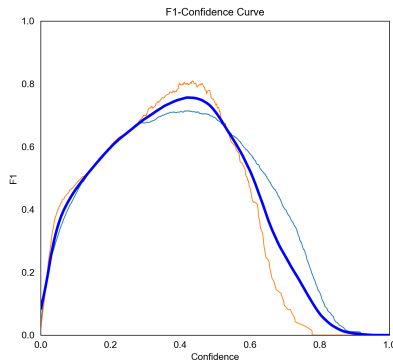
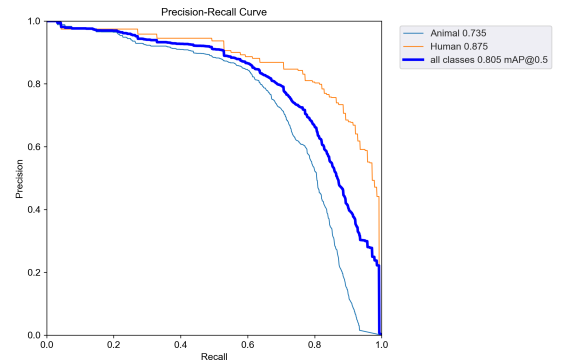


Figure 4: Confusion Matrix of retrained YOLOv5 model

As can be seen in this matrix, the recall of the human class is much improved when the model was retrained on the pseudo-labeled dataset, moving from 39% to 76%. Another note from the confusion matrix is the apparently common prediction of background (or our null class) as animals. We believe that this is a result of bounding boxes formed on inference overlapping with sections of background as well. This is likely due to the data's varied vantage points, which, as mentioned in the Data section, results in varied sizes of animal instances in the training set and therefore, difficulty in finding the correct bounding box size on inference. The vantage point of the images containing human instances seem to be more consistent across the dataset, which may explain why there is less occurrence of human predictions overlapping with the background. We also produced the following visualizations, demonstrating the performance of the model in predicting the two classes



(a) F1-Confidence Curve

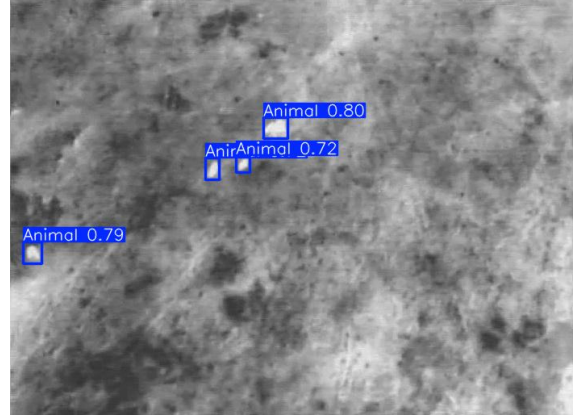


(b) Precision-Recall Curve

The output of this model looks like the images below



(a) Inference on Human Class



(b) Inference on Animal Class

As can be seen in these two images, the model often does not make high-confidence predictions on instances of the human class as a result of a lack of instances in the training set. It can also be seen that the model failed to identify a human on the left side of the image, but was able to identify a majority of the cluster. Another key feature of the model is its ability to perform inference quickly, allowing for the real-time analysis of LWIR video feed.

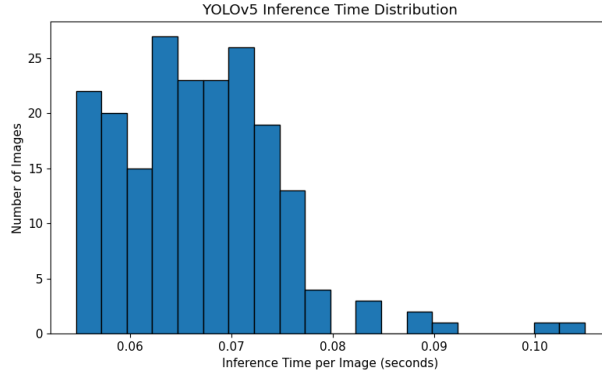


Figure 7: Inference Speed Distribution on Test Set(200 images)

As can be seen in the histogram above, a vast majority of inferences on a single video frame can be performed in less than .1 seconds, with an average inference time of .0669 seconds. This allows for the real-time analysis of 9 Hz LWIR video feed, which is a common frame-rate for this type of camera

5 Discussion

We have designed a computer vision model built off of a pre-trained YOLOv5 that is capable of identifying and differentiating between humans and animals in long-wave infrared images with fairly high success. Through the production of more labeled data and weighted loss, we were able to create a model that is capable of identifying humans at a relatively high rate despite a class imbalance in the dataset. This system has the potential to be applied to real-time LWIR video feed collected by UAVs, allowing for more efficient tracking of human activity in protected wildlife areas and, ideally, reduction in poaching rates.

References

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